

EDUCATION	University of Notre Dame	IN, USA
	<i>Ph.D. in Electrical Engineering</i>	2023 – Present
	• Advisor: Prof. Mengxue Hou. Research: robot manipulation, uncertainty-aware perception, vision-language-action models.	
	Duke University	NC, USA
	<i>M.S. in Mechanical Engineering and Materials Science</i>	2021 – 2023
	• GPA: 3.57/4.00	
	Florida Institute of Technology	FL, USA
	<i>B.S. in Mechanical Engineering</i>	2018 – 2020
	• GPA: 3.78/4.00	

- PUBLICATIONS
1. **Yu Zhou** and Mengxue Hou, “BayesGS: Bayesian Gaussian Splatting for Uncertainty-Aware Real-Time Manipulation,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Late Breaking Results*, 2026.
 2. **Yu Zhou** and Mengxue Hou, “Uncertainty-aware 3D Gaussian Splatting (3DGS) based Pose Estimation for Underwater Vehicles,” in *OCEANS 2025 Great Lakes*, IEEE, pp. 1–6, 2025.
 3. **Yu Zhou**, Ruochu Yang, and Mengxue Hou, “Flow Field Estimation in Underwater Vehicle Navigation using Sporadic Image Observations,” in *International Conference on Ubiquitous Robots (UR)*, 2025.
 4. Likai Pei, **Yu Zhou**, Xingtian Wang, Xueji Zhao, Wanxin Huang, Boyang Cheng, Halid Mulaosmanovic, Stefan Duenkel, Dominik Kleimaier, Sven Beyer, Kai Ni, Mengxue Hou, Michael Niemier, and Ningyuan Cao, “Towards Uncertainty-aware Robotic Perception via Mixed-signal BNN Engine Leveraging Probabilistic Quantum Tunneling,” in *ACM/IEEE Design Automation Conference (DAC)*, 2025.
 5. Ruochu Yang, **Yu Zhou**, Fumin Zhang, and Mengxue Hou, “Interleaved LLM and Motion Planning for Scalable Multi-Object Collection in Large Scene Graphs,” *arXiv preprint arXiv:2507.15782*, 2025.

Under Review

1. **Yu Zhou**, Guoyue Xu, Zhiyuan Li, Ruochu Yang, Mengzhao Jia, Meng Jiang, Tengfei Luo, and Mengxue Hou, “Asking Only When Necessary: Selective Human Intervention for Lab-Protocol-Aware VLA Adaptation,” submitted to *IEEE International Conference on Robotics and Automation (ICRA)*, 2027.

RESEARCH EXPERIENCE	Protocol-Aware Human-in-the-Loop VLA Manipulation	2026.05 – Present
	• Built a framework that decides when a VLA policy should execute, ask a human, or take a correction, using action uncertainty, protocol-violation risk, and a risk predictor learned from human corrections. • Deployed $\pi_{0.5}$ (openpi) on a Kinova Gen3 arm and built the teleoperation data pipeline for fine-tuning. • First-author paper submitted to ICRA 2027.	

Uncertainty-Aware Manipulation with Bayesian 3DGS 2026.01 – Present

- Built a variational 3DGS scene model that outputs expected occupancy and its uncertainty from RGB-D observations.
- Used it in a real-time QP velocity controller with a probabilistic collision cost on a Kinova Gen3 arm (ROS 2, MoveIt, RealSense D435i).
- Presented as a poster at IROS 2026 Late Breaking Results.

Semantic Camera–LiDAR Fusion for 3DGS (Ongoing) 2026 – Present

- Aligning image-based SfM reconstructions with LiDAR point clouds collected by different agents, with no shared pose, through object-level semantic Fused Gromov–Wasserstein (FGW) matching.
- Trained a contrastive image–LiDAR embedding (CLIP image branch) so object clusters from both sensors can be compared, then estimated the similarity transform from the FGW coupling.
- Lowest mean Bi-Chamfer error on three nuScenes scenes (0.133 m vs. 0.207 m for the best baseline). Aligned LiDAR depth is used to supervise 3DGS training. Part of an AFOSR/DEPSCoR project.

Uncertainty-Aware Navigation for Underwater Vehicles 2024 – 2025

- Estimated ocean flow fields from sporadic image observations in HoloOcean simulation (UR 2025).
- Placed variational distributions on 3DGS parameters to estimate pose uncertainty for underwater localization (OCEANS 2025).

INTERNSHIPS **Brain Tool Lab, Duke University | NC, USA** 2022.05 – 2022.07

- Trained a deep learning model to plan laser orientation relative to tissue, minimizing excessive laser-induced ablation.

Fiberhome | Wuhan, China 2020.06 – 2020.08

- Designed main product structures and produced engineering drawings in SolidWorks.

SKILLS

Programming: Python, C++

Robotics: ROS 2, MoveIt, Kinova Gen3, Intel RealSense, HoloOcean

ML / Vision: PyTorch, 3D Gaussian Splatting, COLMAP, VLA fine-tuning (openpi)

CAD / Simulation: SolidWorks, ANSYS Workbench

Languages: Mandarin (native), English (fluent)

TEACHING

Guest Lecturer: EE 67074 Introduction to AI Planning (Fall 2025)

Teaching Assistant: EE 40085 Autonomous Mobile Robots (Fall 2025)

EE 30210 Random Phenomena in EE (Spring 2025)

SERVICE

Reviewer: ICRA, IROS, RO-MAN, Journal of Intelligent & Robotic Systems

Volunteer: IROS 2023

AWARDS

Academic Scholarship, Florida Institute of Technology (GPA top 10%)

2019.02